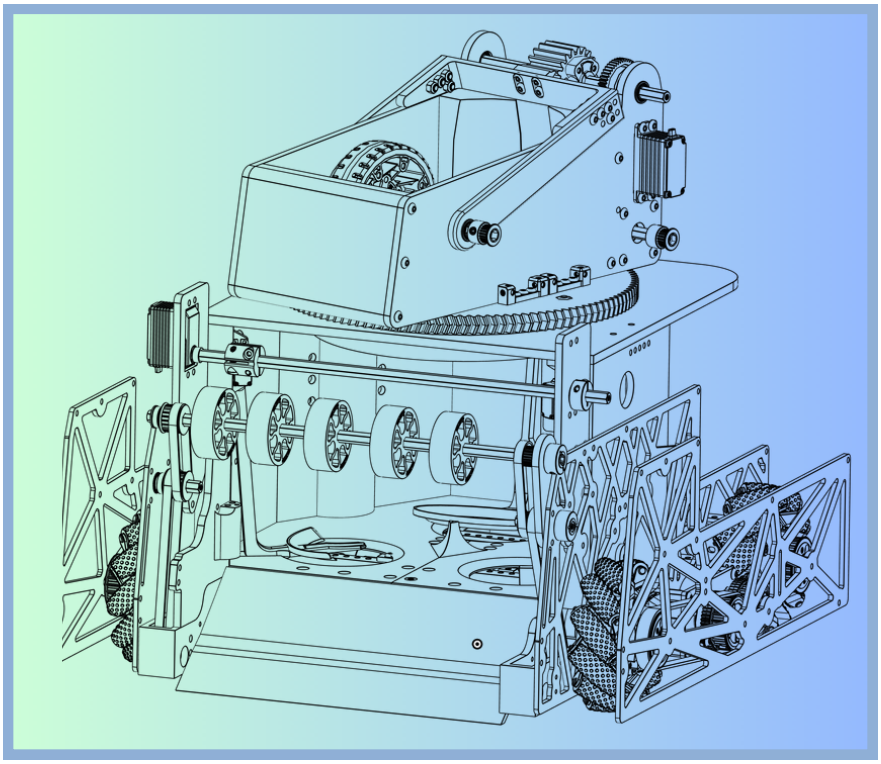
Engineering Portfolio

2025-2026 Season

C.A.R.T
#23280

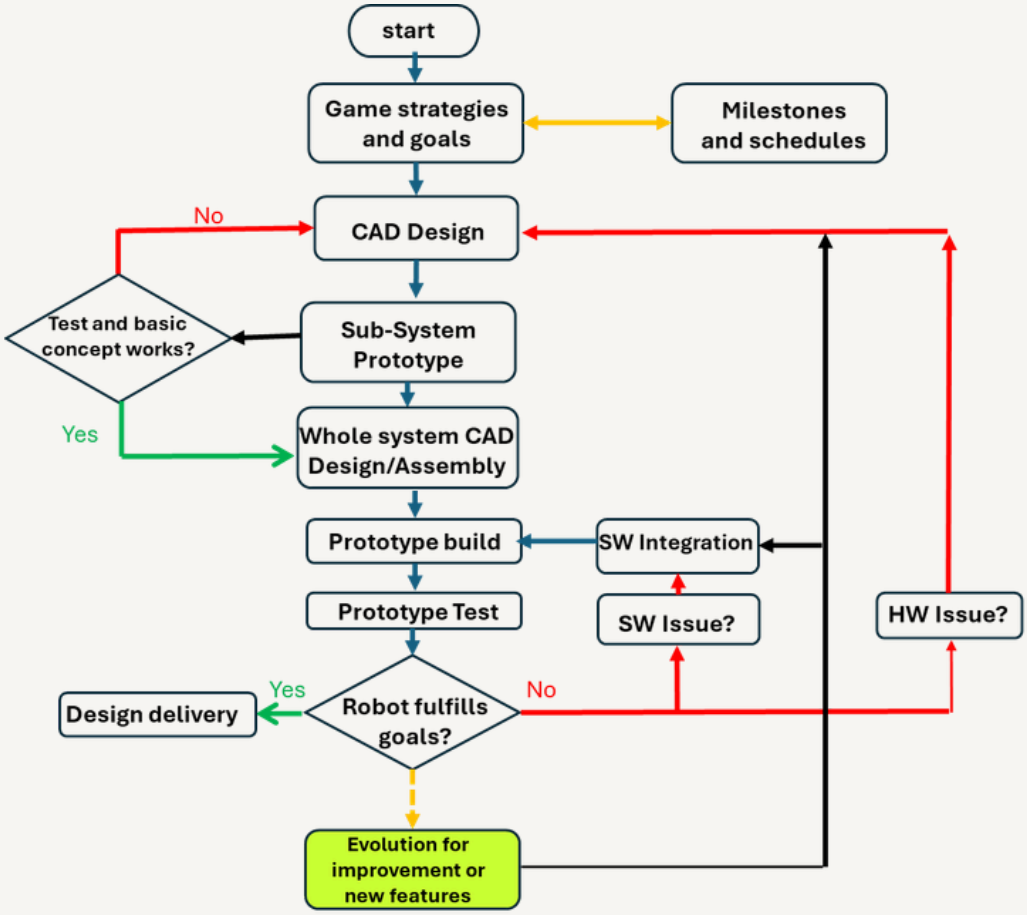


C.A.R.T #23280

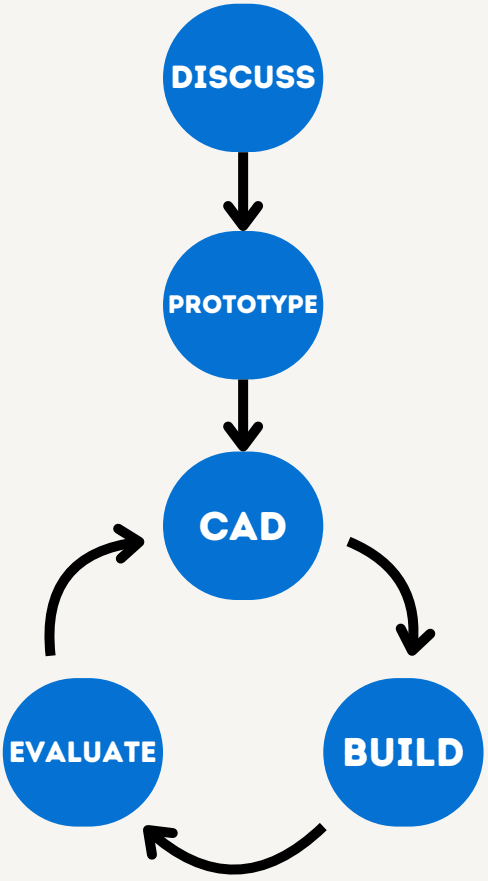
DECODE

DESIGN PROCESS & ROBOT VIEW

Engineering Design Process

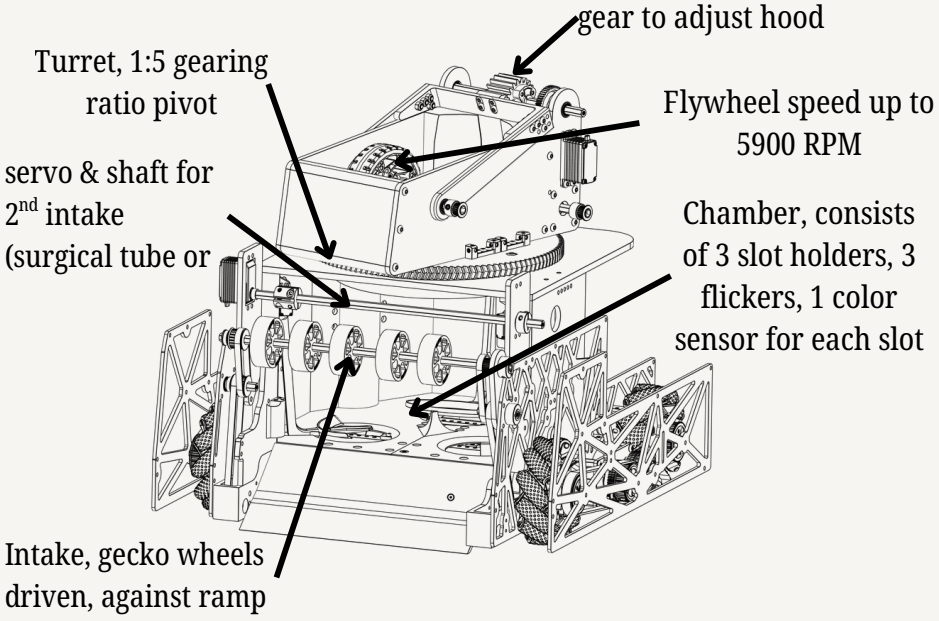


HW Process



Motor, Servo, and Sensor used:

- 4 motors for drivetrain (435 RPM)
- 1 motor for intake (312 RPM)
- 1 motor for turret turning (435RPM)
- 1 motor for flywheel (5800 RPM)
- 3 servos for ball flickers
- 1 servo for 2nd intake
- 1 servo for hood tuning
- 1 limelight camera
- 3 color sensors
- 3 color indicators
- 1 IMU for odometry



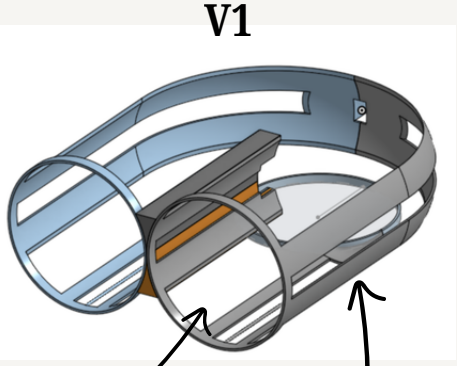
Detachable-plates design makes it easier in repairing and adding components

HARDWARE DESIGN -
Chamber, Flippers, and Evolutions

Chamber Initial Prototypes

- These prototypes were our first iteration of an efficient storage mechanism
- One tube stored the purple artifact and the other stored the green artifact.
- Each artifact organized to be shot in a specific sequence

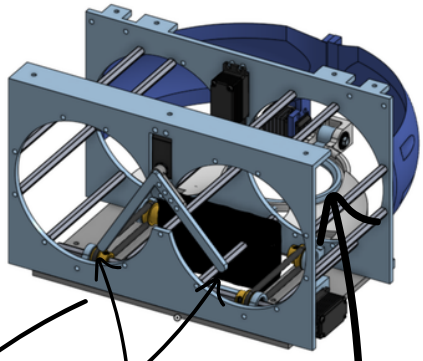
V1



Artifact Entrance

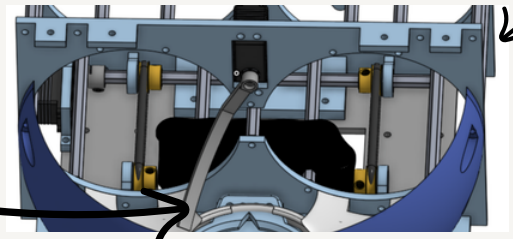
Chamber: Artifact Storage

V2



Entrance gate: hold artifact in

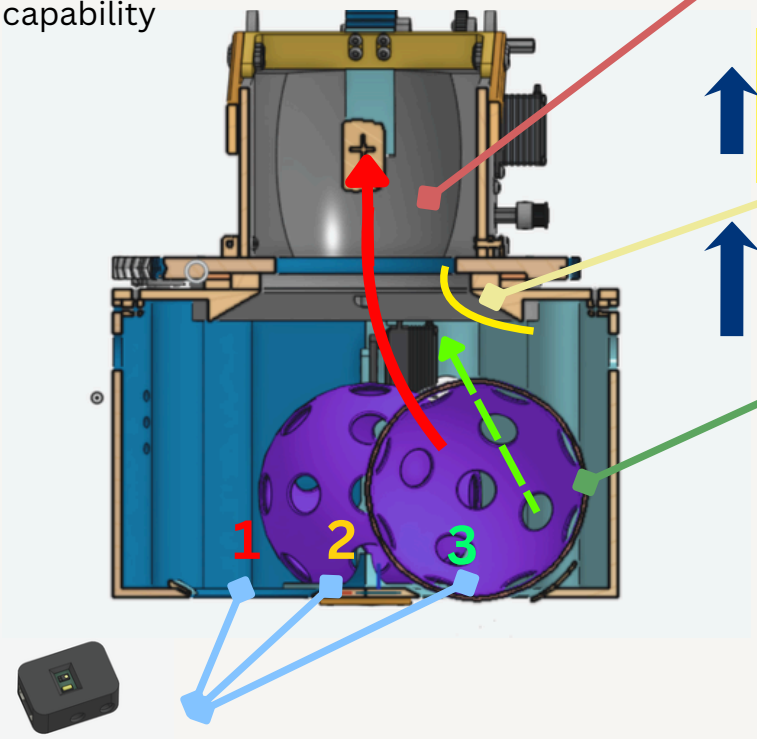
Artifact Flicker: pushes artifact into turret



Artifact Sorter

Chamber Design (V3):

3 independent ball-holding slots for maximum storage capability

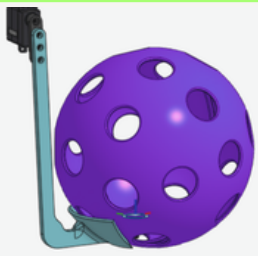


3 color sensors ensuring the ball shooting order with reliability and accuracy.

Result: Smooth transfer from chamber to ball shooting mechanism.

Process: Chamber ceiling directs upward moving ball into central shooter hole.

Action: Flicker hits ball at a 45 degree angle with enough force to push it upwards.



Flicker design with conformal rammer head and axon servo*

Pros:

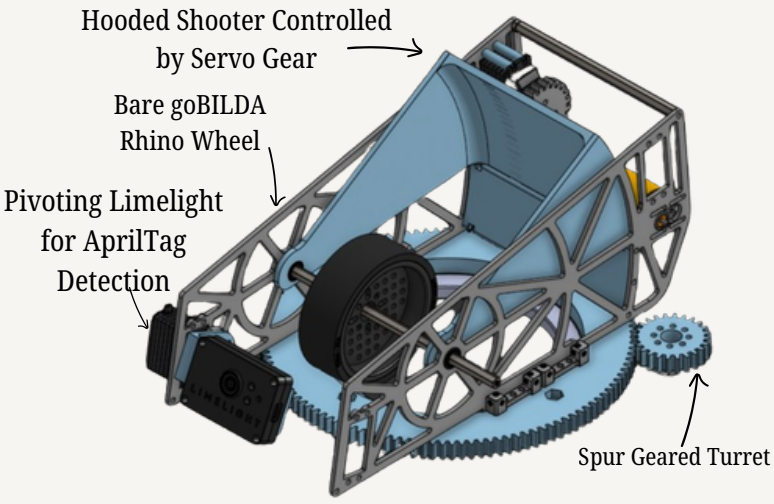
- Independent Ball Control + High Reliability.
- Maximum ball count achieved.
- Ball mobility made possible to combat flicker failures.
- Reduced complexity for ease of use.

Cons:

- Possible flicker failures.
- Drivers must manage all chamber actions.
- May be hard to observe inside layout in some situations.

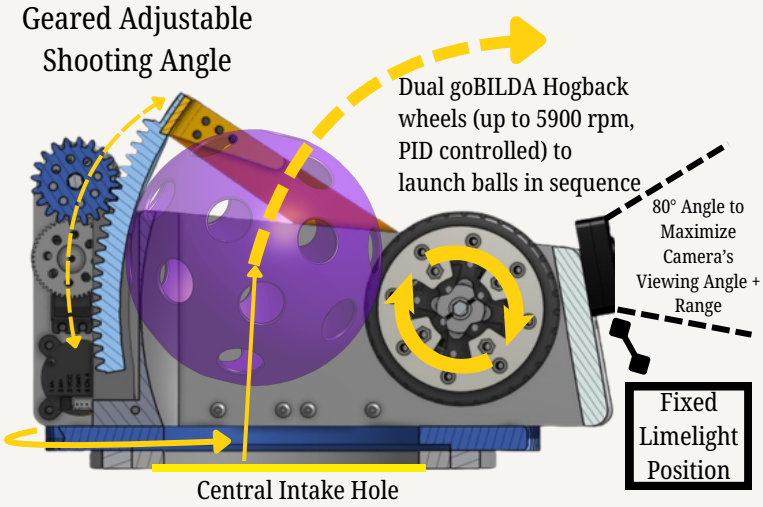
HARDWARE DESIGN -
Intake, Turret, Shooter and Evolutions

Shooter & Turret Design V1



- **First iteration of shooter mechanism**
- **Simple angle adjustment system**
- **COTS lazy-susan bearing for turret rotation**
- **The ARTIFACT enters from the below hole and uses the wheel to compress the ARTIFACT against the hood**

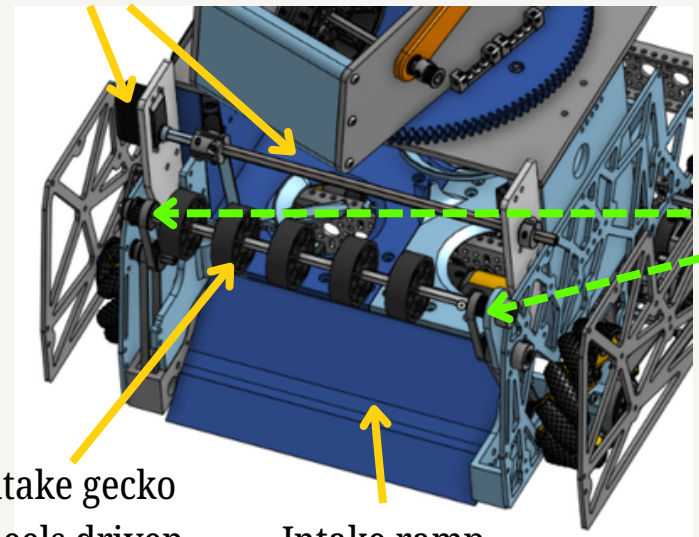
Shooter Design (V2):



- More range from increased firing angle
- Tighter tolerances provides less error
- Lightweight, simple 3D-printed structure
- Ease of repair from modular design
- Multi-ARTIFACT shooting with increased inertia from flywheel weights

Intake Design

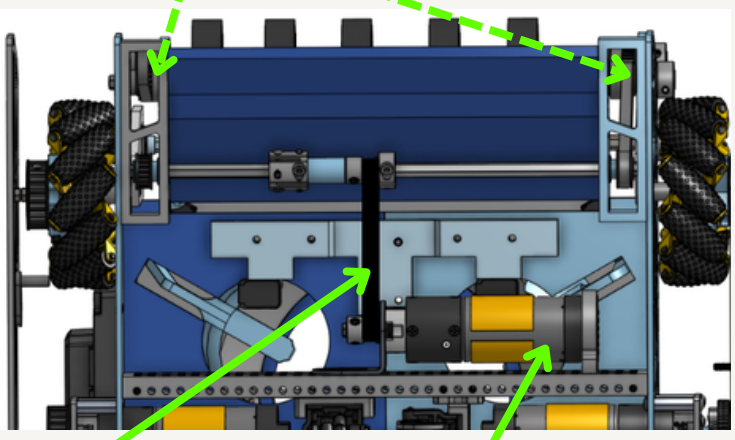
Servo and Shaft for the 2nd intake (surgical tube or gecko wheel)



Intake gecko wheels driven, against ramp

Intake ramp

Pulleys and belts at the ends of the intake shaft to transfer balanced speed and power to the gecko wheels for stable and fast ball intaking



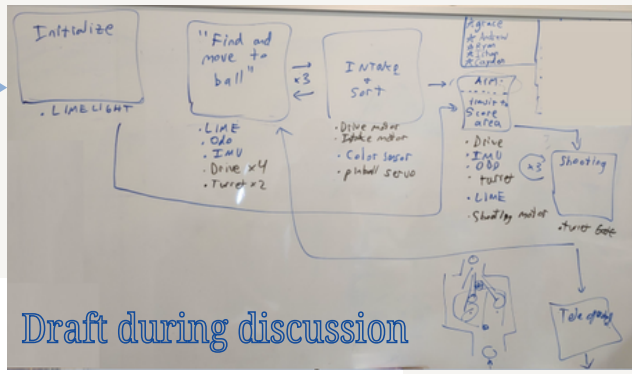
Belt

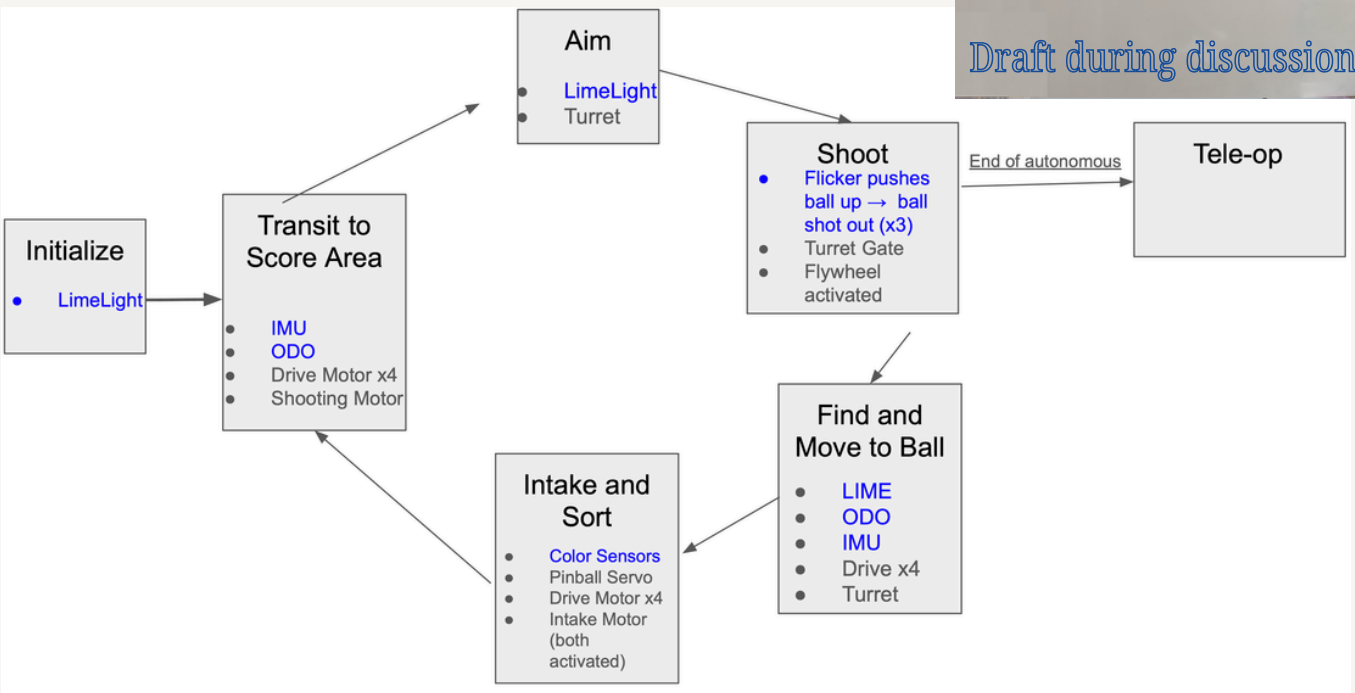
Single motor (312 RPM) to transfer speed and power to the main intake

*****A lot of experiments were done to determine ramp slope and its surface friction for ball intaking**

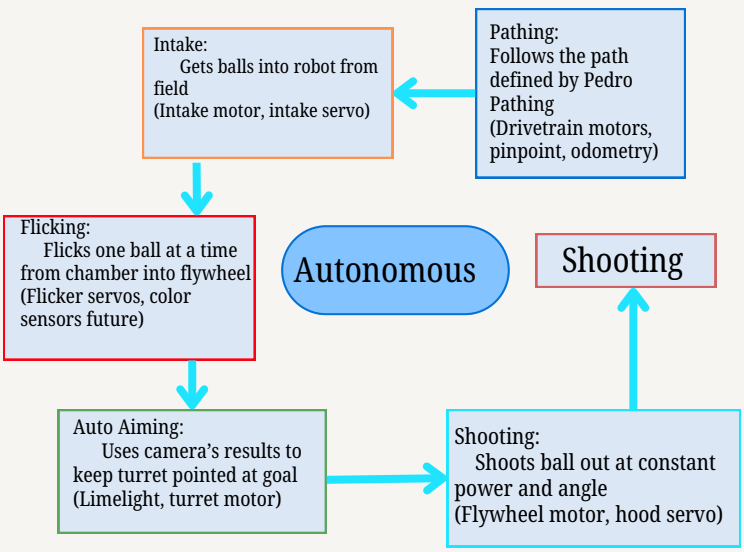
SOFTWARE DESIGN -
State Machines & Strategies for Auto and Teleop

OVERALL

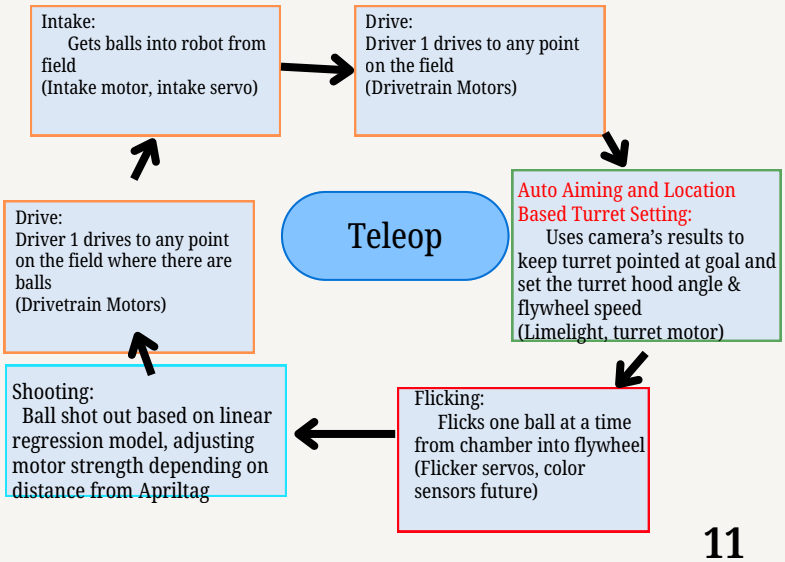




AUTO STRATEGY



TELEOP STRATEGY



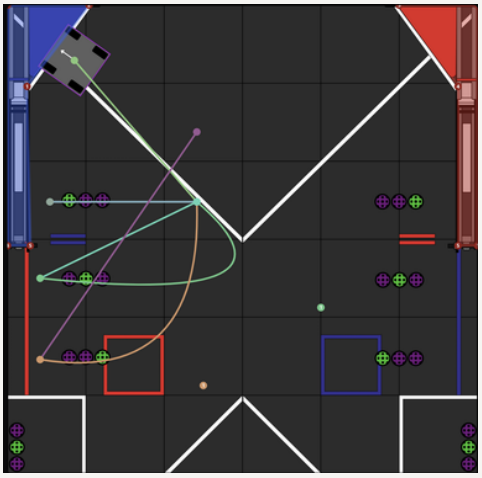
SOFTWARE DESIGN

PEDRO PATHING FOR MOTION PLANNING

Instead of roadrunner, we use Pedro Pathing for robot trajectory planning

- 12 ball trajectory & 6 ball far trajectory on both sides
- Passive odometry with IMU fusion sensor
- Automatic tuning:
  - Tunes speed and deceleration
- Apply PID Tuning to the changes of Drive, Heading, Translational, and Centripetal

Blue Auto Tragectory



SENSORS FOR PERFORMANCE

LimeLight Camera

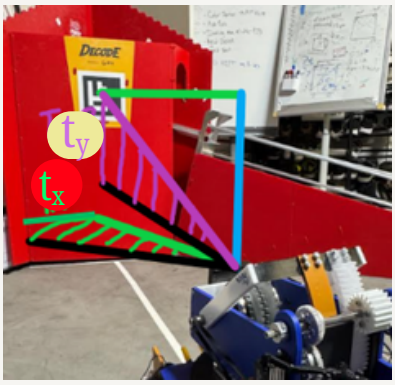
- Function: Turret aiming for ball shooting and April Tag detection
- Return Values
  - vertical offset angle, t_y, and the horizontal offset angle, t_x
    - Values used to move the turret accordingly
    - Used for controlling hood angle and shooting power

What we have done:

Use PID implementation

- to regulate turret aiming at apriltags
- Uses return values mentioned previously to calculate movements

- derivative_t_x = 1000000000.0 * (limelight_t_x - last_t_x) / (System.nanoTime())
  - Makes the turret movement smooth and stops the turret from over-shooting
- pPID * limelight_t_x + dPID * derivative_t_x
  - set power for motor (smooth and efficient)
- No Integral (I) term because P&D terms can make the turret consistently lock on the target



t_x and t_y from camera view



Color Sensor and Experiments

- We use 3 color sensors to detect the color of balls in the chamber to shoot balls following the pattern on the obelisk
- We did experiment to compare different sensors for color sensor decision-making
- We decided to use Rev color sensor based on the experiment results

Vendor	Detect Principle	Plug & Play	Peformance Variation	Impact from Light change	unit by unit	successful rate at 2cm (10 times)
Brushland lab sensor	RGB photodiodes	Need setting at I2C and change to analog port	big	big	not quiet consistent	70%
Rev color sensor	RGB + Alpha (clear) channel detection	yes	very small	small	very consistent	100%



IMU From Odometry Computer

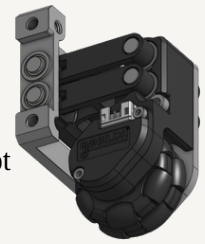
- IMU is used in odometry for auto trajectory (please refer to Advanced Mathematics for Passive Odometry)

RGB Indicators

- 3 RGB Indicators are used to illustrate the color of the ball in each ball holder in the chamber based on color sensor detection so shooting

ADVANCED MATHEMATICS

ODOMETRY AND MATHEMATICS APPLIED



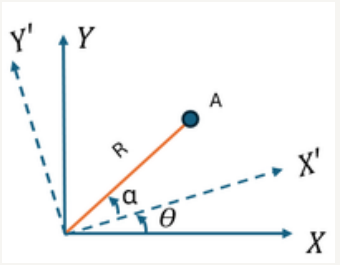


Fig 1. The coordinates for field & robot

- No-powered wheels are used to estimate the position (x, y) and orientation (Θ) of the robot in the case of drivetrain wheels slipperage.

Assume XY is the field coordinate system, and X' Y' is the robot coordinate system which rotates Θ degree (Fig 1) . A point A can be presented as:

$$\begin{matrix} X = R * \cos(\alpha + \Theta) \\ Y = R * sin(\alpha + \Theta) \end{matrix} \text{ and } \begin{matrix} X' = R * \cos(\alpha) \\ Y' = R * sin(\alpha) \end{matrix} \rightarrow \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \cos(\Theta) & -\sin(\Theta) \\ \sin(\Theta) & \cos(\Theta) \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix}$$

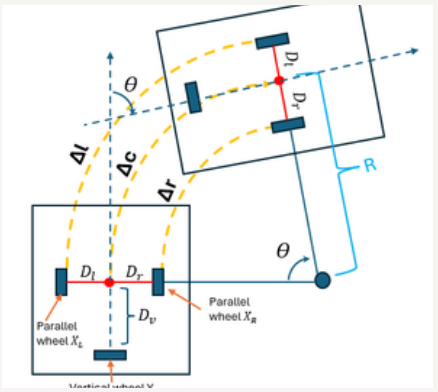


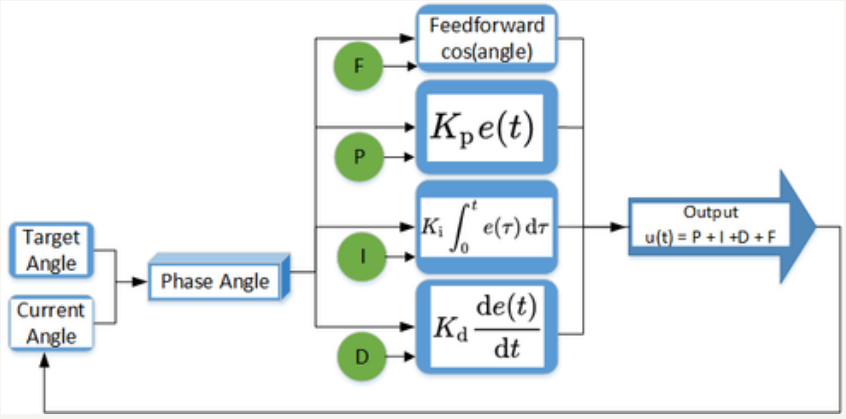
Fig 2. Robot moves with heading turn of Θ

Δl, Δc, and Δr are the displacement of left parallel wheel, center of the robot, and the right parallel wheel, apply Arc=R* Θ to them, we have $\Theta = \frac{\Delta l - \Delta r}{D_l + D_r}$ and $\Delta c = \frac{\Delta l + \Delta r}{2}$

If the data sampling of the encoder is high enough, the displacement of the vertical wheel is Δv, the rotation angle at the current sample is θ, we will have the robot displacement (Δx, Δy) along x, y axis from the following matrix relations.

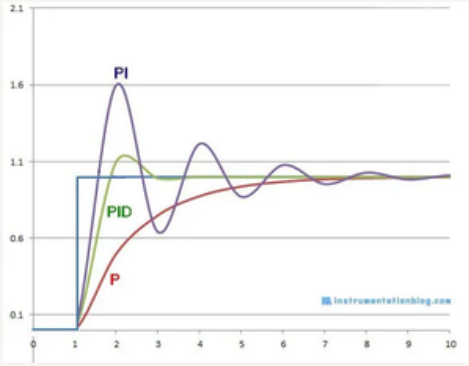
$$\begin{bmatrix} \Delta x \\ \Delta y \\ \theta \end{bmatrix} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} \Delta c \\ \Delta v \\ \theta \end{bmatrix} = \begin{bmatrix} \Delta c * \cos(\theta) - \Delta v * \sin(\theta) \\ \Delta c * \sin(\theta) + \Delta v * \cos(\theta) \\ \theta \end{bmatrix}$$

PID (+FEEDFORWARD) CONTROL LOOP



PID Control are used at the following scenarios in our design:

1. Motion Trajectory Planning in Auto for accurate drivetrain locations
2. Turret Aiming to the goal basket
3. Flywheel speed

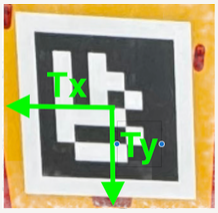


PI is the slope that stops the oscillation as the p is shown getting converging to a flat line at 0

INNOVATION--
Adaptive Guided Shooting With Distance

Our Goals: To created an algorithm that automatically adjusts the shooter and shoot speed based on the robot position in the field

Benefits: Robot will shoot balls autonomously and accurately, independent of robot defense driving, strategical moving, and collisions from other robots.



The limelight gave us the vertical offset degree t_y - how far above or below the center of camera view the target is. we obtain two data sets, for the optimal hood angle and flywheel speed that corresponded to the t_y.

t_y		
−7.2	0.55	0.85
−6.2	0.55	0.85
−5.2	0.55	0.95
−4.2	0.575	0.95
−3.2	0.60	0.90
−2.2	0.60	0.90
−1.2	0.625	0.85
0.2	0.65	0.90
1.2	0.65	0.85
2.2	0.65	0.95

Data sampling

We applied **least-squares linear regression** to model the relationship between t_y and shooter

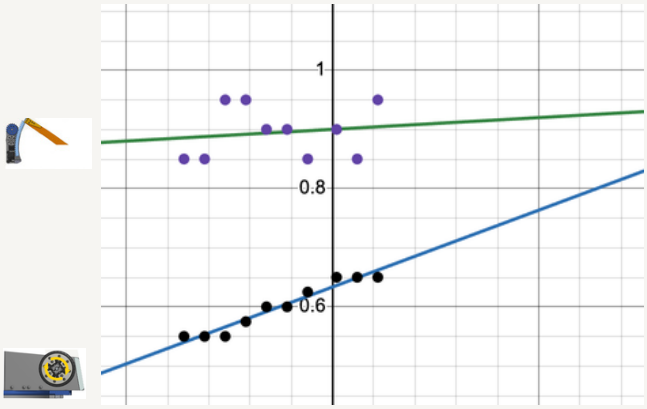
The formula (x=ty)

Predicting f(x) for hood angle

$$f(x) = 0.0129883x + 0.63351$$

Predicting a(x) for flywheel speed

$$a(x) = 0.00198387x + 0.900118$$



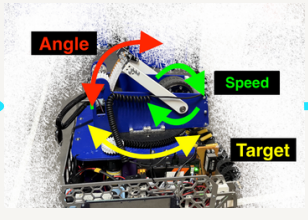
Predicting of the angle and speed

How It Works:

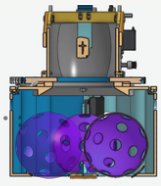
1. The camera detects an AprilTag on the field.
2. The turret uses vision data to automatically align with the target.
3. The turret uses the predictive model to automatically adjust the hood angle and flywheel speed based on the robot's position.
4. Sensor data tracks how many balls the robot has in chamber.
5. The robot's odometry is calculated using wheel encoders and the IMU to provide field-relative pose (x, y, heading).
6. The shooter shoots automatically when the target is aligned and the system is ready.



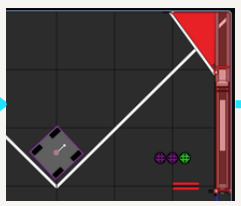
1. Detect



2. Align & 3. Adjust



4: Balls in chamber



5. In launch Zone



6. Goal